

Interrupt

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Interrupt

What is an interrupt?

An interrupt is a signal from a device attached to a computer or from a program within the computer that causes the CPU to stop its normal program execution and perform service related to the event.

Examples of interrupts :I/O completion, divide-by-0, etc.

Maskable Interrupt: It is a hardware interrupt that may be ignored by setting a bit in an interrupt mask register's (IMR) bit-mask.

Nonmaskable Interrupt: is a hardware interrupt that does not have a bit-mask associated with it - meaning that it can never be ignored. NMIs are often used for timers, especially watchdog timers

Interrupt Service Routine

A complete interrupt service cycle includes

1. Saving the program counter value in the stack
2. Saving the CPU status (including the CPU status register and some other registers) in the stack
3. Identifying the cause of interrupt
4. Resolving the starting address of the corresponding interrupt service routine
5. Executing the interrupt service routine
6. Restoring the CPU status and the program counter from the stack
7. Restarting the interrupted program (continue)

Interrupt Service Routine

An interrupt handler, also known as an interrupt service routine (ISR), is a callback subroutine in an operating system whose execution is triggered by the reception of an interrupt. Interrupt handlers have a multitude of functions, which vary based on the reason the interrupt was generated.

Interrupt Overhead

- The interrupt overhead is caused by context switching (storing and restoring the state of CPU)
- On interrupt handler entry, the context of the current process and its thread must be saved. On exit, it must be restored.
- On handler entry, memory locations different from the memory locations in the cache are used, and therefore cache updates are required.

Interrupt nesting:

-An interrupt can happen while executing an ISR. This is called *interrupt nesting*.

